

SPHERE: MISSION CONTROL

TEACHER PEDAGOGICAL DEPLOYMENT MANUAL

DOCUMENT REF // EU_SPHERE_INSTRUCTOR_V2

Instructor Overview: This document maps the SPHERE Thermal Suit experiment to standard European secondary physics frameworks. It includes lesson timings, safety protocols, expected analytical constants, and a complete grading rubric to streamline execution in single-session classroom windows.

1. CURRICULAR ALIGNMENT & LESSON SCHEME

This curriculum is aligned with the **Inquiry-Based Science Education (IBSE)** pedagogy, specifically mimicking the virtual-physical calibration workflows of the **European Union Horizon 2020 SPHERE initiative**. It directly maps to secondary physics curriculum segments covering:

- **Thermodynamics:** Thermal conduction, convection, radiation, thermal barriers, and specific heat capacity calculations.
- **Mathematical Modeling:** Rate equations, Newtonian decay curves, and parametric modeling.
- **IoT & Digital Twins:** Real-time sensory inputs mapped onto a digital math engine to create predictive twins.

THE 45-MINUTE LESSON TIMELINE BLUEPRINT:

TIMING	INQUIRY STEP	STUDENT ACTIVITIES & MILESTONES
00 - 10 Min	Engage & Certify	Students review the deep-space context, read the student briefing, and complete the Pre-Flight Quiz to unlock mission telemetry.
10 - 20 Min	Explore & Construct	Student crews wrap their physical PET jars with specific insulation configs and thread thermal wire probes through PG7 glands.
20 - 25 Min	Model & Predict	Crews open the Simulator tab, select their insulation schemes, configure parameters ($T_0 = 80^{\circ}\text{C}$), and execute mathematical predictions.
25 - 40 Min	Explain & Measure	Pour 80°C water into PET jars. Submerge in the 0°C ice baths. Start stopwatch. Students log temperatures every 60 seconds.
40 - 45 Min	Elaborate & Grade	Analyze the resulting Thermal Correlation Score . Discuss convective leaks, and record grading coefficients.

2. RISK ASSESSMENT & HAZARD MITIGATION

Enforce these standard safety mandates to guarantee zero accident rates during live laboratory sessions:

HAZARD	LEVEL	MANDATED MITIGATION STEPS
Hot Water Scalding	HIGH	Do not allow students to use water > 80°C. Teachers should pre-boil and dispense water pre-mixed with cold tap water to lock thermal limits at exactly 80°C.
Melted Ice Slippages	MEDIUM	Place physical ice baths in high-density plastic trays. Provide moisture absorbent towels. Keep ice bath tables completely segregated from laptops and tablets.
Electrical Shock	MEDIUM	Only permit low-voltage, USB-powered or battery-powered digital sensor readouts. Keep the thermocouple cables long to maximize distance from the computer screen.
Capsule Deformation	LOW	Ensure thin PET food jars are not exposed to boiling 100°C temperatures which causes polymer warping. Enforce the strict 80°C limit.

3. ANALYTICAL SOLUTIONS KEY (NEWTONIAN DECAY CONSTANTS)

Crews configure their spacesuits with distinct insulation wraps. Use this analytical key to verify if their physical curves map correctly to physical heat coefficients:

EXPECTED COOLING RATES AFTER 15-MINUTE SUBMERSION:

INSULATION DESIGN	DECAY CONSTANT (k)	EXPECTED END TEMP (15 MIN)	DOMINANT COOLING PATHWAY BLOCKED
A. Bare Capsule	$k \approx 0.082$	$\approx 23^{\circ}\text{C}$	Baseline Scientific Control. No pathways blocked.
B. Cotton Padding	$k \approx 0.045$	$\approx 40^{\circ}\text{C}$	Blocks direct thermal conduction through fibers.
C. Bubble Wrap Rings	$k \approx 0.032$	$\approx 49^{\circ}\text{C}$	Blocks convective thermal transfer via static air pockets.
D. Multi-Layer MLI	$k \approx 0.015$	$\approx 64^{\circ}\text{C}$	Blocks radiant energy transfer via reflective aluminum layers.

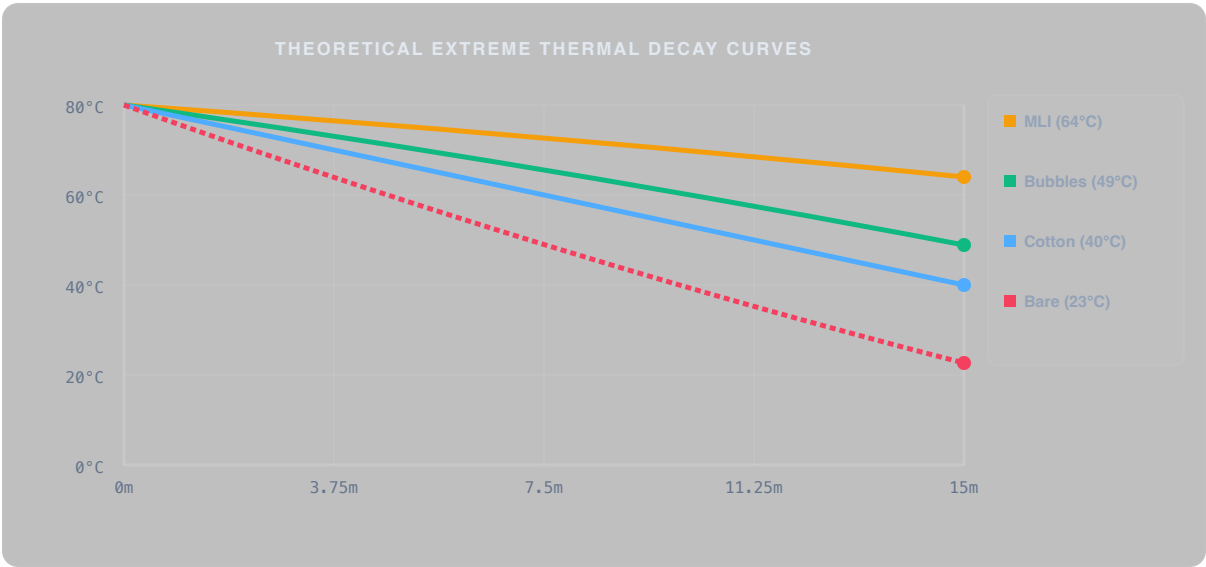


FIGURE 3: EXPECTED EXPONENTIAL COOLING CURVES (t = 0 to 15m)

*Note: Physical values may deviate by $\pm 5\%$ depending on wrapping tightness and capsule sealing. The real-time **Thermal Correlation Score** computed in the telemetry dashboard will rate how closely they match these digital curves.*

4. CLASSROOM EVALUATION & GRADING RUBRIC

Use this standardized 100-point rubric to grade student performance:

GRADING CRITERIA	POINTS	EXCELLENT (100% Points)	SATISFACTORY (70% Points)	INSUFFICIENT (30% Points)
1. Assembly Precision	20 Pt	Capsule sealed completely. PG7 gland tightened perfectly. DS18B20 wire is centered without touching PET walls.	Capsule sealed, but slight water leakage inside jar lid. Probe touching side wall.	Severe leaks. Wire gland is loose or untightened. Complete water ingress.
2. Telemetry Diligence	20 Pt	All 15 data points entered precisely every 60 seconds. Active team tracking.	Missed 2-3 data points. Data recorded late or values estimated.	Missed major telemetry slots. Incomplete logging.
3. Correlation Score	30 Pt	Physical curve matches mathematical simulator curve with $> 90\%$ correlation score.	Thermal correlation falls between 75% and 89%. Small wrapping air gaps.	Correlation score is $< 70\%$ due to major physical leaks or simulation error.
4. Analytical Report	30 Pt	Deep explanation of conduction, convection, and radiation blocks. Precise calculations.	Answers standard questions correctly but lacks deep scientific rationale.	Incomplete answers. Scientific concepts misunderstood or omitted.